🪐 Exoplanet Habitability: A Data Science Exploration

Can we estimate how likely a known exoplanet is to be habitable — using only its orbital and host-star parameters? And can a machine learning model learn that pattern well enough to generalize?

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Overview

- **Data**: 6,372 confirmed exoplanets pulled live from the NASA Exoplanet Archive (TAP API), cleaned down to 5,578.
- **Feature engineering**: Habitable Zone boundaries (Kopparapu-inspired) and the full 4-parameter Earth Similarity Index (radius, density, escape velocity, temperature).
- **Result**: 12 planets meet all three habitability criteria (Habitable Zone + rocky radius + ESI > 0.7) — nearly all match well-known literature picks (Proxima Cen b, TRAPPIST-1e, TOI-700d, Kepler-442b).
- **Machine learning**: a tuned Random Forest predicts the habitability heuristic from raw stellar/orbital parameters alone (PR-AUC 0.48 vs. ~0.002 random baseline), led by stellar mass, radius, and temperature.

Three things we caught (and fixed) along the way

1. **The Radius Valley** — our 1.6 R⊕ rocky-planet cutoff landed almost exactly on a well-known gap in the real exoplanet size distribution, confirming the threshold wasn't arbitrary.
2. **An ESI temperature bug** — some planets (e.g. Kepler-438 b) scored a near-perfect ESI while sitting outside the Habitable Zone. The cause: comparing planet temperatures to Earth's greenhouse-warmed surface temperature (288K) instead of its blackbody equilibrium temperature (255K). Fixing the reference realigned ESI with the Habitable Zone.
3. **Detection bias in `sy_dist`** — our 12 candidates sat ~8x closer to Earth than the dataset average. Not physics — nearby planets are simply easier to measure precisely. Removing this feature from the model *improved* performance (PR-AUC 0.471 → 0.480).

Repository structure

```
.
├── notebooks/
│   └── exoplanet_habitability_analysis.ipynb   # Full analysis, step by step
├── presentation/
│   ├── exoplanet_habitability_presentation.html
│   └── exoplanet_habitability_presentation.pdf
├── requirements.txt
└── README.md
```

Running it yourself

bash

```
pip install -r requirements.txt
jupyter notebook notebooks/exoplanet_habitability_analysis.ipynb
```

The notebook fetches live data from the NASA Exoplanet Archive — no API key needed.

Limitations

- Only 12 positive examples — any model evaluation at this scale carries real statistical uncertainty.
- The habitability label is a simplified rule-based heuristic, not a scientific determination — it ignores atmosphere, magnetic field, tidal locking, and moons entirely.
- The underlying catalog is biased toward planets that are easy to detect (large, close-in) and easy to characterize precisely (nearby systems).

Data source

This research has made use of the NASA Exoplanet Archive, which is operated by the California Institute of Technology, under contract with NASA under the Exoplanet Exploration Program.